# B1.2 Proteins

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-b1-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 210–222 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

## From sequence to shape

### B1.2.6 R-groups

pp. 211–212

All amino acids share an amine group, a carboxyl group and a hydrogen on one carbon; they differ in their **R-group**. R-groups are **non-polar** (hydrophobic) or hydrophilic; hydrophilic ones are **polar** or **charged**, either acidic (negative) or basic (positive). Twenty R-groups in any order give an almost endless variety of proteins, because the R-groups decide how a chain folds and what it binds.

### B1.2.7 Primary structure determines shape

pp. 212–213

The **primary structure** is the sequence of amino acids, set by the gene. The sequence fixes where every R-group sits, so it determines the three-dimensional shape. That is why proteins have precise, repeatable structures, and why small unfolded proteins can refold correctly on their own.

### B1.2.8 Secondary structure

pp. 213–214

Parts of the chain coil into an **alpha helix** or lie side by side as a **beta-pleated sheet**. Both are stabilized by **hydrogen bonds at regular positions** between the C=O of one amino acid and the N-H of another in the backbone, not between R-groups.

# Folding and assembly

## B1.2.9 Bonds in tertiary structure

pp. 214–216

### Tertiary structure: bonds between R-groups

B1.2.9

| Bond | Between | Strength and sensitivity |
|---|---|---|
| Hydrogen bonds | Polar R-groups | Weak; broken by heat |
| Ionic bonds | + amine and − carboxyl R-groups | Moderate; changed by pH |
| Disulfide bonds | -SH groups of two cysteines | Strong covalent |
| Hydrophobic interactions | Non-polar R-groups clustered away from water | Weak alone; many together |

**Charge depends on pH**
Amines gain $H^+$ and carboxyls lose $H^+$, so pH changes ionic bonds and can denature proteins.

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Tertiary structure is held by bonds between R-groups: **hydrogen bonds** between polar groups, **ionic bonds** between charged groups, **disulfide bonds** between pairs of cysteines, and **hydrophobic interactions** among non-polar groups. Amine and carboxyl groups gain or lose $H^+$, so their charge, and the ionic bonds, depend on pH.

## B1.2.10 Hydrophobic and hydrophilic amino acids

pp. 216–217

In soluble globular proteins, non-polar amino acids cluster in the **core** and polar ones face the water. Integral membrane proteins have **non-polar regions** where they cross the lipid bilayer and polar regions where they meet water; channel proteins also have polar amino acids lining the pore.

## B1.2.11 Quaternary structure

pp. 217–220

Some proteins have two or more polypeptides. **Insulin** (two chains joined by disulfide bonds) and **collagen** (three chains in a triple helix) are non-conjugated. **Haemoglobin** has four chains, each with a **haem** prosthetic group containing iron, so it is a conjugated protein.

## B1.2.12 Globular and fibrous proteins

pp. 220–222

**Globular** proteins such as insulin are compact and soluble; insulin's precise shape lets it bind its receptor as it travels in the blood. **Fibrous** proteins such as collagen are long and insoluble; collagen's repeating sequence, with glycine at every third position, lets three chains pack into a strong rope, giving tendons and skin tensile strength.

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